

LogiCORE IP Integrated Bit Error Ratio Tester (IBERT) for 7 Series GTP Transceivers v3.0

***Product Guide for Vivado
Design Suite***

PG133 March 20, 2013

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Introduction

The customizable LogiCORE™ IP Integrated Bit Error Ratio Tester (IBERT) core for 7 series FPGA GTP transceivers is designed for evaluating and monitoring the GTP transceivers. This core includes pattern generators and checkers that are implemented in FPGA logic, and access to ports and the dynamic reconfiguration port attributes of the GTP transceivers.

Communication logic is also included to allow the design to be run time accessible through JTAG. This core can be used as a self-contained or open design, based on customer configuration, and as described in this document.

Features

- Provides a communication path between the Vivado™ serial I/O analyzer feature and the IBERT core
- Provides a user-selectable number of 7 series FPGA GTP transceivers
- Transceivers can be customized for the desired line rate, reference clock rate and reference clock source
- Requires a system clock that can be sourced from a pin or one of the enabled GTP transceivers

LogiCORE IP Facts Table	
Core Specifics	
Supported Device Family ⁽¹⁾	Artix™-7
Supported User Interfaces	N/A
Resources	See Table 2-1 and Table 2-2 .
Provided with Core	
Design Files	RTL
Example Design	Verilog
Test Bench	Not Provided
Constraints File	XDC
Simulation Model	Not Provided
Supported S/W Driver	N/A
Tested Design Flows ⁽²⁾	
Design Entry	Vivado™ Design Suite
Simulation	Not Provided
Synthesis	Vivado Synthesis
Support	
Provided by Xilinx @ www.xilinx.com/support	

Notes:

1. For a complete listing of supported devices, see the Vivado IP catalog.
2. For the supported versions of the tools, see the [Xilinx Design Tools: Release Notes Guide](#).

Overview

Functional Description

The IBERT core provides a broad-based Physical Medium Attachment (PMA) evaluation and demonstration platform for 7 series FPGA GTP transceivers. Parameterizable to use different GTP transceivers and clocking topologies, the IBERT core can also be customized to use different line rates and reference clock rates. Data pattern generators and checkers are included for each GTP transceiver desired, giving several different Pseudo-random binary sequence (PRBS) and clock patterns to be sent over the channels.

In addition, the configuration and tuning of the GTP transceivers is accessible through logic that communicates to the Dynamic Reconfiguration Port (DRP) port of the GTP transceiver, in order to change attribute settings, as well as registers that control the values on the ports. At run time, the Vivado™ serial I/O analyzer communicates to the IBERT core through JTAG, using the Xilinx cables and proprietary logic that is part of the IBERT core.

Feature Summary

The IBERT core is designed for PMA evaluation and demonstration. All the major PMA features of the GTP transceiver are supported and controllable, including:

- TX pre-emphasis and post-emphasis
- TX differential swing
- RX equalization
- Phase-Locked Loop (PLL) divider settings

Some of the Physical Coding Sublayer (PCS) features offered by the transceiver are outside the scope of IBERT, including:

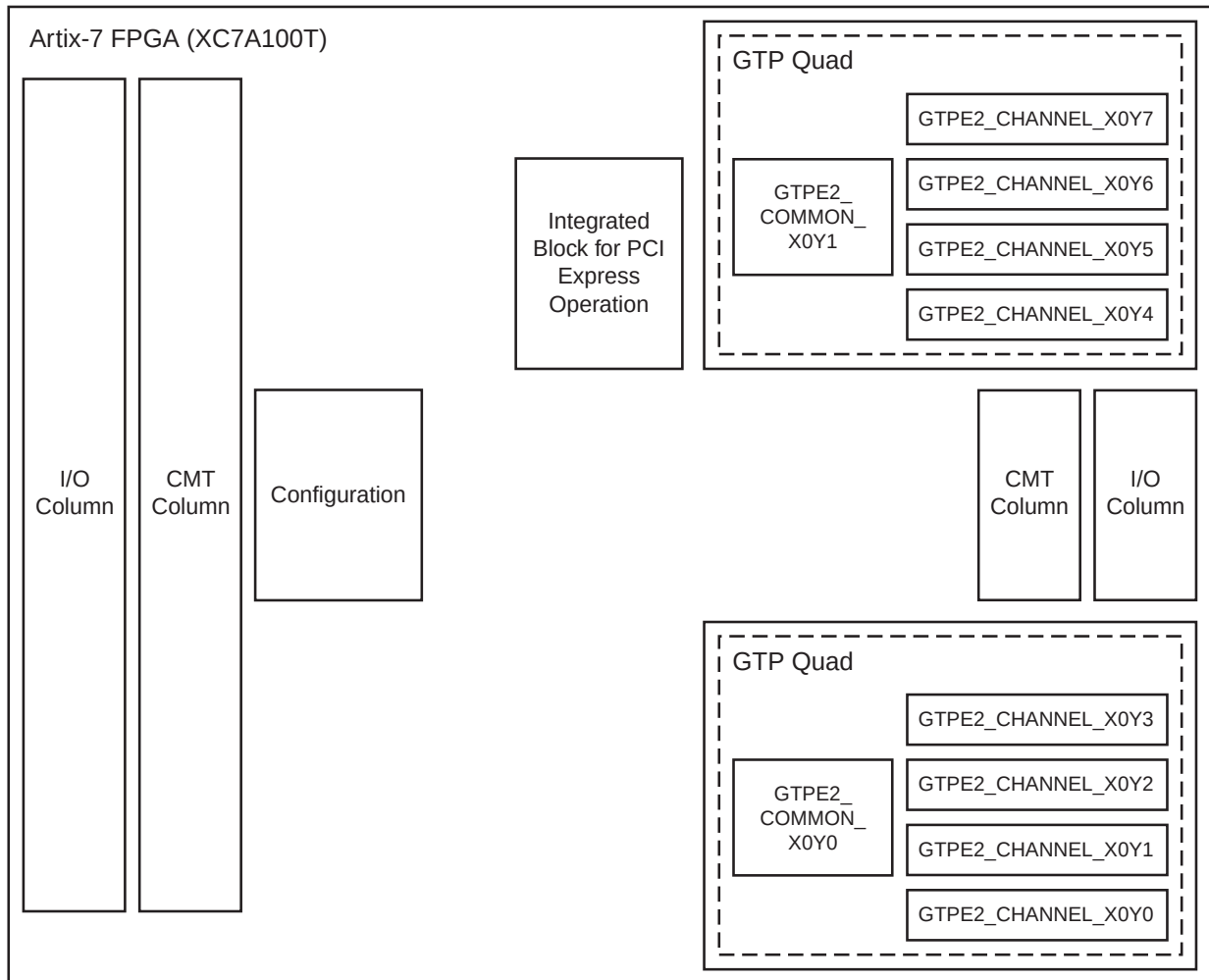
- Clock Correction
- Channel Bonding
- 8B/10B, 64B/66B, or 64B/67B encoding

- TX or RX Buffer Bypass

PLL Configuration

For each Quad (4 channels) there are two shared ring PLLs (PLL0 and PLL1). Any of the 4 channels in the Quad can use either PLL.

Figure 1-1 shows a Quad in a 7 series device. The GTPE2_CHANNEL component has the serial transceiver and the GTPE2_COMMON has PLL0 and PLL1.



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Figure 1-1: Quad in a 7 Series Device

The serial transceiver REFCLK can be sourced from either PLL0 or PLL1 based on multiplexers shown in Figure 1-2.

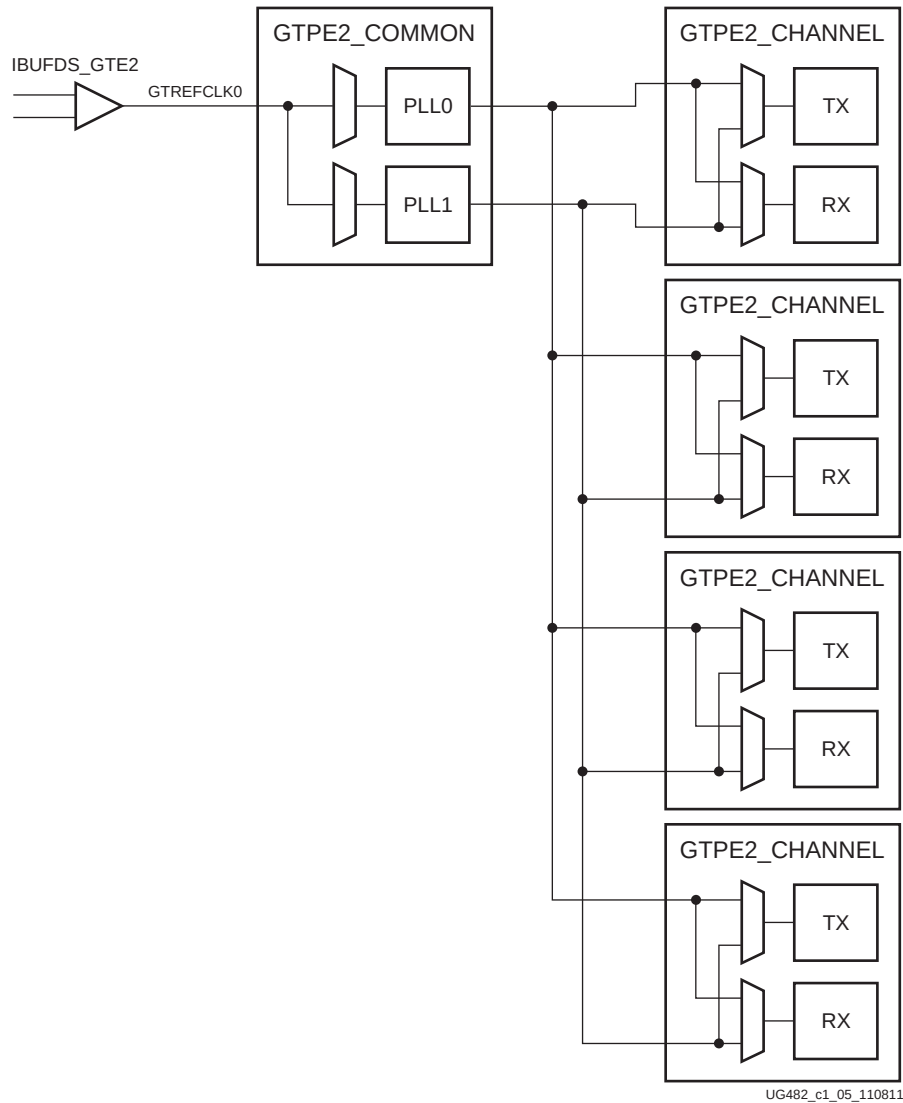


Figure 1-2: Serial Transceiver REFCLK Sourcing

Pattern Generation and Checking

Each GTP transceiver enabled in the IBERT design has a pattern generator and a pattern checker. The pattern generator sends data out through the transmitter. The pattern checker accepts data through the receiver and checks it against an internally generated pattern. IBERT offers PRBS 7-bit, PRBS 15-bit, PRBS 23-bit, PRBS 31-bit, Clk 2x (101010...), and Clk 10x (11111111110000000000...) patterns.

Using the pattern checker logic, the incoming data is compared against a pattern that is internally generated. When the checker receives five consecutive cycles of data with no errors, the `LINK` signal is asserted. If the `LINK` signal is asserted and the checker receives five consecutive cycles with data errors, the `LINK` signal is deasserted. Internal counters accumulate the number of words and errors received.

DRP and Port Access

You can change GTP transceiver ports and attributes. The DRP interface logic allows the run time software to monitor and change any attribute of the GTP transceivers and the corresponding PLL. When applicable, readable and writable registers are also included that are connected to the various ports of the GTP transceiver. All are accessible at run time using the Vivado serial I/O analyzer.

Applications

The IBERT core is designed to be used in any application that requires verification or evaluation of 7 series FPGA GTP transceivers.

Licensing and Ordering Information

This Xilinx LogiCORE IP module is provided at no additional cost with the Xilinx Vivado Design Suite tool under the terms of the [Xilinx End User License](#).

Information about this and other Xilinx LogiCORE IP modules is available at the [Xilinx Intellectual Property](#) page. For information on pricing and availability of other Xilinx LogiCORE IP modules and tools, contact your [local Xilinx sales representative](#).

Product Specification

Performance

The IBERT core can be configured to run any of the allowable line rates for the GTP transceivers. See the *7 Series FPGAs Overview* [Ref 2] for the line rates supported by speed grade.

Maximum Frequencies

The IBERT core can operate at the maximum user clock frequencies for the fabric width/speed grade selected. The maximum system clock rate is 125 MHz and the generated design divides any incoming system clock to adhere to this constraint.

Resource Utilization

Resources required for the IBERT 7 series GTP transceiver core have been estimated for the Artix™-7 FPGA (Table 2-1 and Table 2-2). These values were generated using Vivado™ IP Catalog. They are derived from post-synthesis reports, and might change during place and route.

Table 2-1: Configuration Details

Configuration	Device	IBERT Setup
Config1	XC7A100T-FGG676-3	Line rate: 1.25 Gb/s; Refclk Frequency: 125 MHz; No. of Quads: 1.
Config2	XC7A100T-FGG676-3	Line rate: 3.125 Gb/s, Refclk Frequency: 125 MHz; No. of Quads: 2.
Config3	XC7A100T-FGG676-3	Line rate: 6.6 Gb/s; Refclk Frequency: 165 MHz; No. of Quads: 2.

Table 2-2: Artix-7 FPGA Resource Estimates

Configuration	Device Resources				Performance
	LUTs	Flip-Flops	DSP Slices	Block RAMs	F _{Max} (MHz)
Config1	6,597	12,442	0	0	236,432
Config2	12,784	24,296	0	0	272,066
Config3	12,792	24,304	0	0	272,066

Port Descriptions

The I/O signals of the IBERT core consist only of the GTP transceiver reference clocks, the GTP transceiver transmit and receive pins, and a system clock (optional).

Table 2-3: IBERT I/O Signals

Signal Name	I/O	Description
SYSCLOCK_I	I	Clock that clocks all communication logic. This port is present only when an external clock is selected in the generator.
TXN_0[7:0]	O	Transmit differential pairs for each of the n GTP transceivers used.
TXP_0[7:0]		
RXN_0[7:0]	I	Receive differential pairs for each of the n GTP transceivers used.
RXP_0[7:0]		
GTREFCLK0_I[1:0]	I	GTP transceiver reference clocks used. The number of MGTREFCLK ports can be equal to or less than the number of transmit and receive ports because some GTP transceivers can share clock inputs.
GTREFCLK1_I[1:0]		
RXOUTCLK_0	O	Quad based RX output clock.

Designing with the Core

This chapter includes guidelines and additional information to facilitate designing with the core.

General Design Guidelines

GTP Transceiver Naming Style

There are two conventions for naming the GTP transceiver, based on the location in the serial transceiver tile in the device. M and n in XmYn naming convention indicate the X and Y coordinates of the serial transceiver location. M and n in serial transceiver m_n naming convention indicating serial transceiver number and quad associated.

Line Rate Support

IBERT supports a maximum of three different line rates in a single design. For each of these line rates, you can select a custom value based on your requirements, or you can choose from pre-provided industry standard protocols (for example, CPRI™, Gigabit Ethernet, or XAUI). Specify the number of serial transceivers for each line rate that is programmed with these settings. Because usage of QPLL is recommended for line rates above 6.5 Gb/s, you can select QPLL/CPLL for each line rate falling in the range 0.6 Gb/s to 6.5 Gb/s.

Serial Transceiver Location

Based on the total number of serial transceivers selected, provide the specific location of each serial transceiver that you intend to use. The region shown in the panel indicates the location of serial transceivers in the tile. This demarcation of region is based on the physical placement of serial transceivers with respect to median of BUFGs available for each device.

Clocking

System Clock

The IBERT core requires a free-running system clock for communication and other logic that is included in the core. This clock can be chosen at generation time to originate from an FPGA pin, or from a dedicated REFCLK input of one of the GTP transceivers. In order for the core to operate properly, this system clock source must remain operational and stable when the FPGA is configured with the IBERT core design.

If the system clock is running faster than 150 MHz, it is divided down internally using an Mixed-Mode Clock Manager (MMCM) to satisfy timing constraints. The clock source selected must be stable and free running after the FPGA is configured with the IBERT design. The system clock is used for core communication and as a reference for system measurements. Therefore, the clock source selected must remain operational and stable when using the IBERT core.

Receiver Output Clock

The receiver clock probe enable is provided to pull out a recovered clock from any serial transceiver, if desired. When enabled, a new panel appears just before the summary page where you can fill in the serial transceiver source and probe pin standards.

Reference Clock

The reference clock source should be provided for all the serial transceivers selected. The drop-down list provides you with possible sources based on local clocks in the same quad and shared clocks from north/south quads.

Resets

Run time resets are available for the BERT counters and all GT resets are available.

Customizing and Generating the Core

This chapter includes information about using Xilinx tools to customize and generate the core in the Vivado™ Design Suite environment.

Vivado Integrated Design Environment (IDE)

The Vivado IP catalog allows you to define and generate a customized IBERT core to use to validate the transceivers of the device. You can customize the number of serial transceivers, line rate and reference clock, and PLL selection for each serial transceiver.

The IBERT core can be found in /Debug & Verification/Debug/ in the Vivado IP Catalog.

To access the core name, perform the following:

1. Open a project by selecting **File** then **Open Project** or create a new project by selecting **File** then **New Project** in Vivado.
2. Open the IP catalog and navigate to any of the taxonomies.
3. Double-click **IBERT 7 Series GTP** to bring up the IBERT Customize IP dialog box.

[Figure 4-1](#) to [Figure 4-4](#) show the IBERT Customize IP dialog boxes with information about customizing ports.

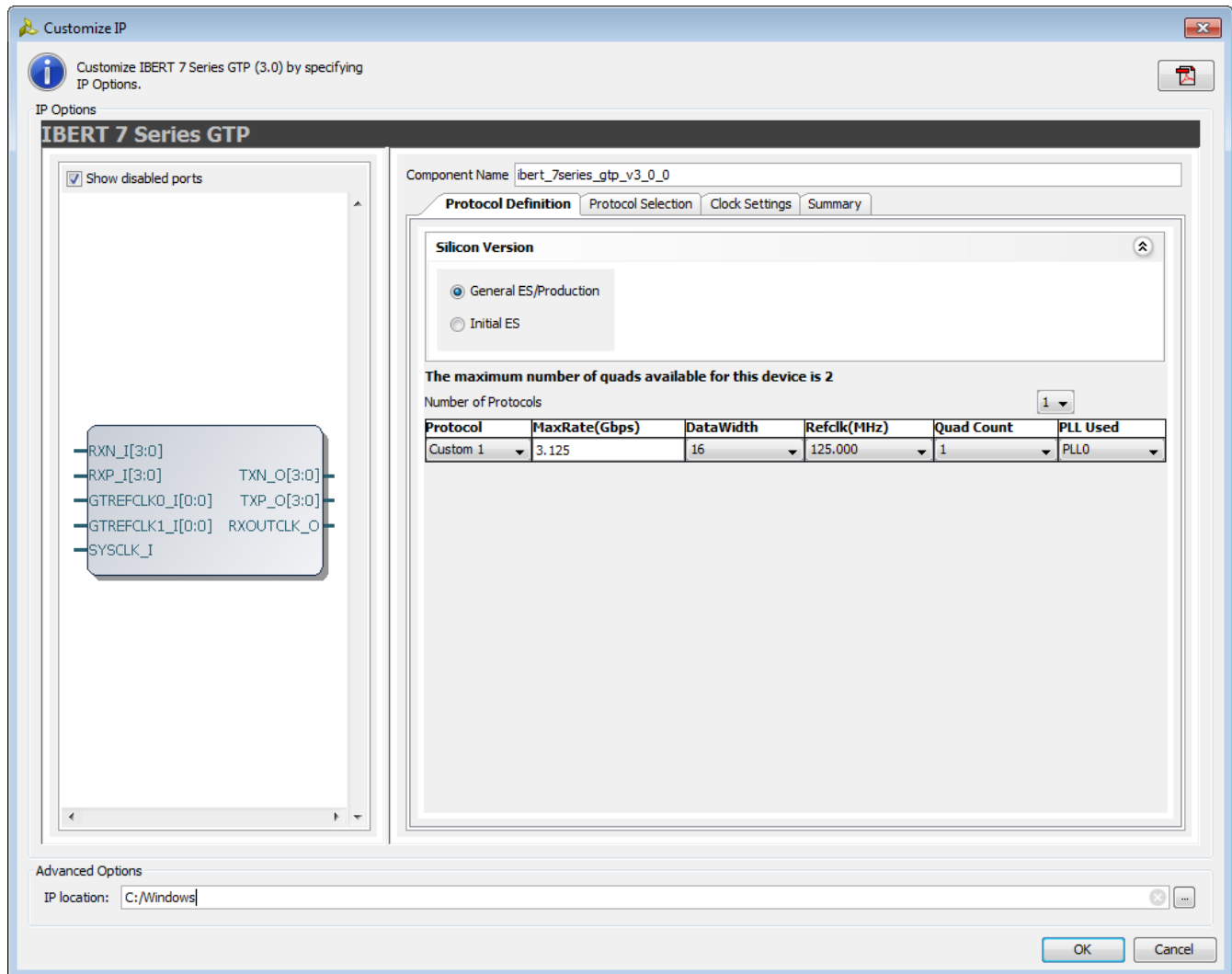


Figure 4-1: Vivado Customize IP Dialog Box – Protocol Definition

Entering the Component Name

The Component Name field can consist of any combination of alpha-numeric characters including the underscore symbol. However, the underscore symbol cannot be the first character in the component name.

Silicon Version

Choose either the **General ES/Production** or **Initial ES** radio button to match the silicon targeted.

Protocol Definition

A protocol is a line rate/data width/reference clock rate combination. Up to three protocols can be defined for an IBERT core, and any number of available Quads can be designated as any protocol defined.

1. Choose the number of protocols desired.
2. In the Protocol combination box, select either Custom or a Pre-defined protocol. If Custom, type in the line rate (the rate appears in red text if outside the range allowed).
 - a. Select the data width.
 - b. Choose the REFCLK rate and the number of Quads running at this rate.
3. Changing the line rate entered changes the choices in the REFCLK combination box.

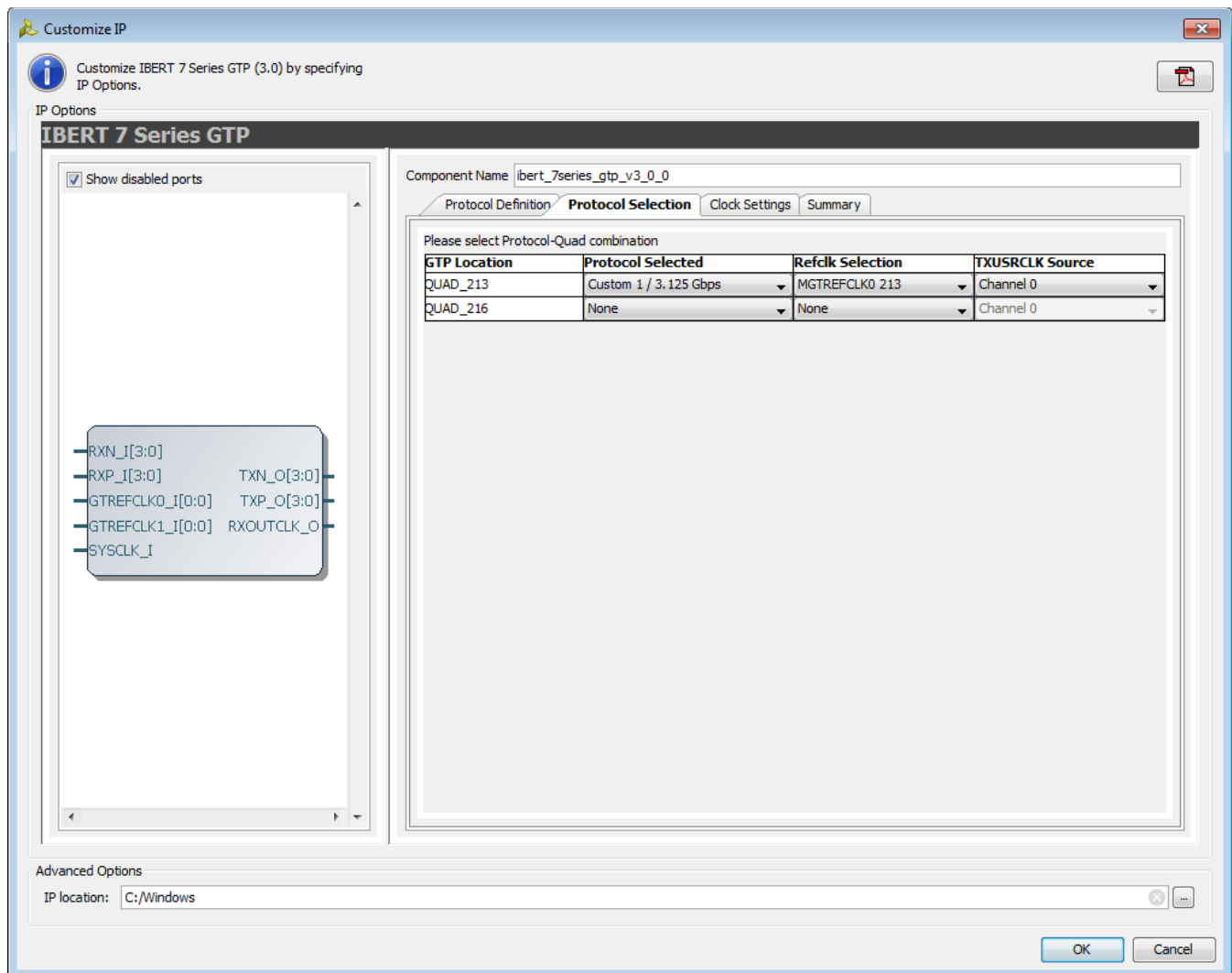


Figure 4-2: Vivado Customize IP Dialog Box – Protocol Selection

Protocol Selection

In the **Protocol Selection** tab, the Quads available in the device/package combination are shown. To allocate a Quad to a specific protocol, select it in the **Protocol Selected** combination. The legal choices for the reference clock input are listed in the REFCLK selection combination. After a Quad is allocated to a protocol, the TXUSRCLK source combination is enabled. The TXUSRCLKs are shared among all four channels in the Quad. Use this combination to select which of the four lanes TXOUTCLK should be used for the user clocks.

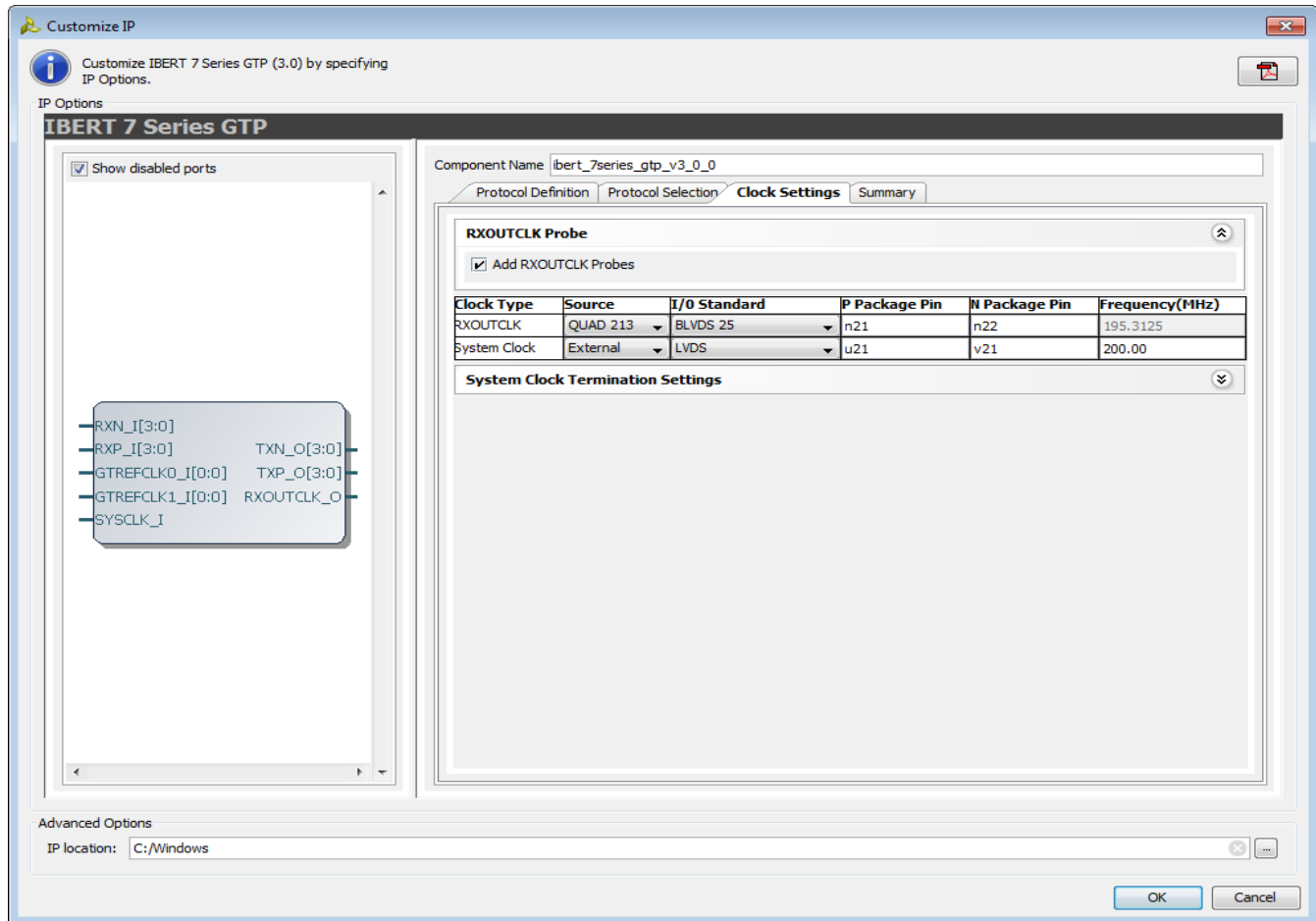


Figure 4-3: Vivado Customize IP Dialog Box – Clock Settings

Clock Settings

In the **Clock Settings** tab, click the **Add RXOUTCLK Probes** to drive an output pin or pin pair with the RXOUTCLK of lane 0 of the Quad. Select the **I/O Standard** from the list and assign valid pin locations.



RECOMMENDED: For the system clock, select an external clock source to drive the system clock of the IBERT design.

Select an **I/O Standard**, valid pin locations, and frequency to complete the system clock settings. Alternatively, the Quad reference clock can be chosen instead.

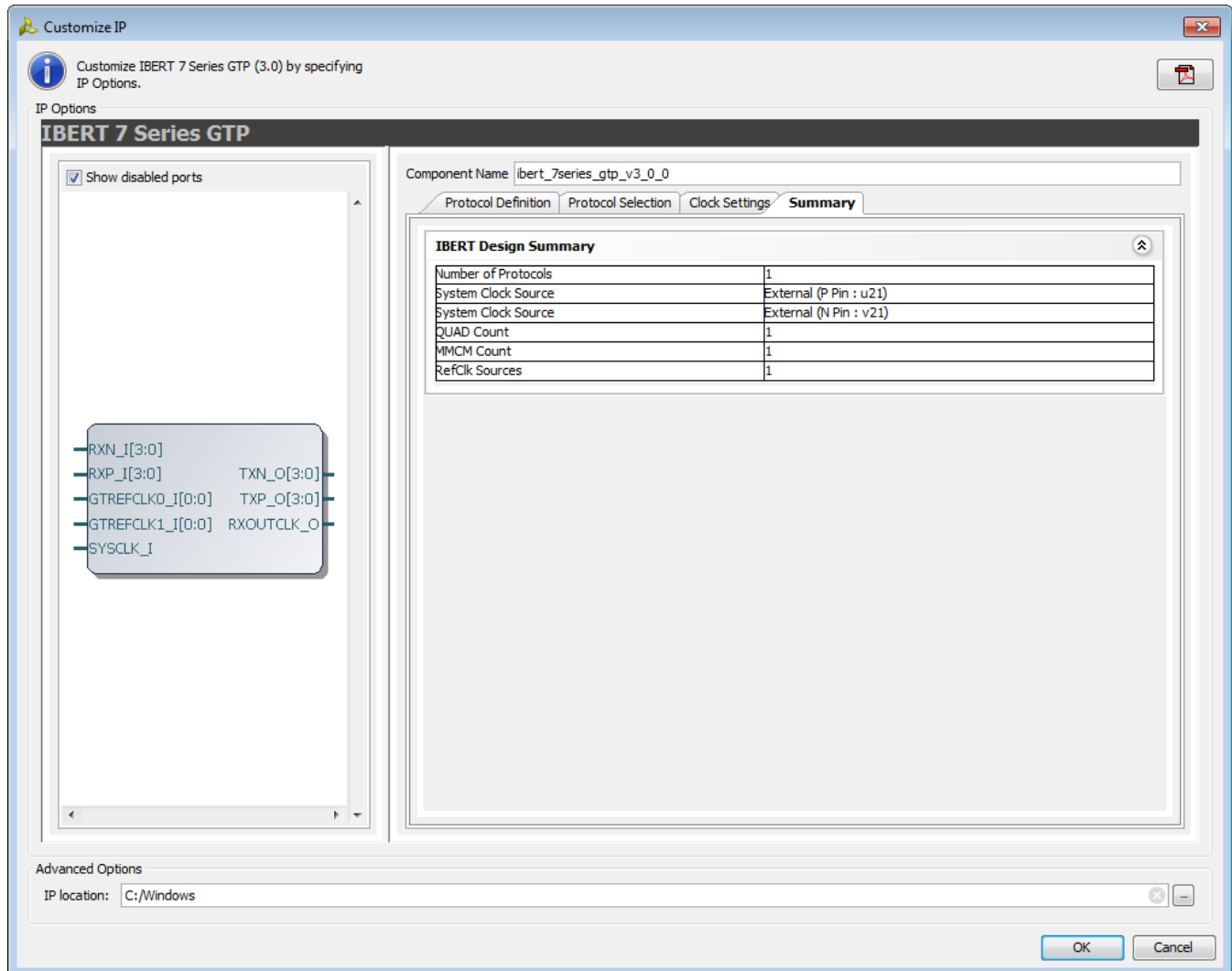


Figure 4-4: Vivado Customize IP Dialog Box – Summary

Summary

Review the settings chosen in the summary page and if they are satisfactory, click **OK** to generate the IBERT core.

Output Generation

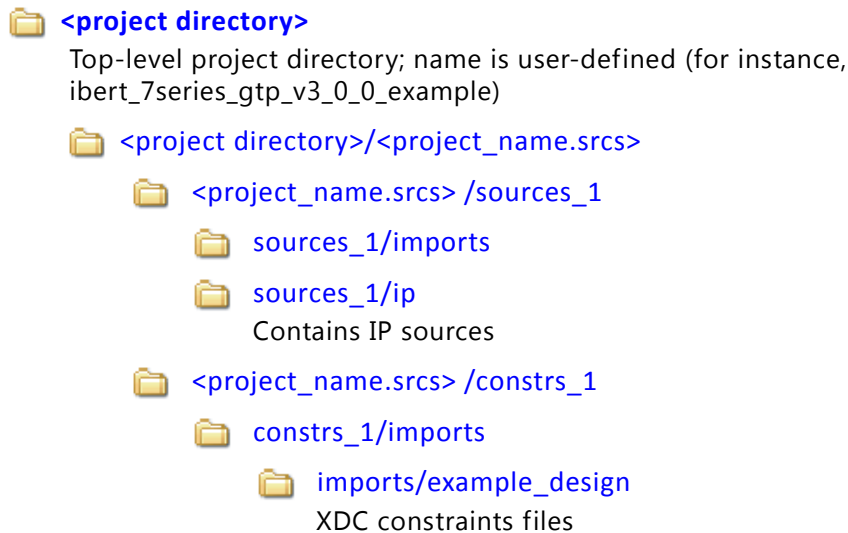
This section provides detailed information about the files and the directory structure generated by the Xilinx Vivado tool.

The output files generated from the Xilinx Vivado IP catalog are placed in the project directory. The file output list might include some or all of the following files.

Generating an Example Design

Once the IP has been generated, right-click the **IP** in the **Sources** view and choose **Open IP Example Design**. This creates a new project with the IBERT example design as the top-level.

Generating the Core



Constraining the Core

This chapter contains information about constraining the core in the Vivado™ Design Suite environment.

Required Constraints

The IBERT GTP core is generated with its own timing and location constraints, based on the choices the user made when customizing the core. No additional constraints are required.

Debugging

This appendix includes details about resources available on the Xilinx Support website and debugging tools. In addition, this appendix provides a step-by-step debugging process to guide you through debugging the IBERT 7 series GTP transceivers core.

The following topics are included in this appendix:

- [Finding Help on Xilinx.com](#)
- [Debug Tools](#)

Finding Help on Xilinx.com

To help in the design and debug process when using the IBERT 7 series GTP transceivers, the [Xilinx Support web page](http://www.xilinx.com/support) (www.xilinx.com/support) contains key resources such as product documentation, release notes, answer records, information about known issues, and links for opening a Technical Support WebCase.

Documentation

This product guide is the main document associated with the IBERT 7 series GTP transceivers. This guide, along with documentation related to all products that aid in the design process, can be found on the Xilinx Support web page (www.xilinx.com/support) or by using the Xilinx Documentation Navigator.

Download the Xilinx Documentation Navigator from the Design Tools tab on the Downloads page (www.xilinx.com/download). For more information about this tool and the features available, open the online help after installation.

Known Issues

Answer Records include information about commonly encountered problems, helpful information on how to resolve these problems, and any known issues with a Xilinx product. Answer Records are created and maintained daily ensuring that users have access to the most accurate information available.

Answer Records for this core are listed below, and can also be located by using the Search Support box on the main [Xilinx support web page](#). To maximize your search results, use proper keywords such as

- Product name
- Tool message(s)
- Summary of the issue encountered

Master Answer Record for the IBERT 7 Series GTP Transceivers

AR [54607](#).

Contacting Technical Support

Xilinx provides technical support at www.xilinx.com/support for this LogiCORE™ IP product when used as described in the product documentation. Xilinx cannot guarantee timing, functionality, or support of product if implemented in devices that are not defined in the documentation, if customized beyond that allowed in the product documentation, or if changes are made to any section of the design labeled DO NOT MODIFY.

To contact Xilinx Technical Support:

1. Navigate to www.xilinx.com/support.
2. Open a WebCase by selecting the [WebCase](#) link located under Support Quick Links.

When opening a WebCase, include:

- Target FPGA including package and speed grade.
- All applicable Xilinx Design Tools and simulator software versions.
- Additional files based on the specific issue might also be required. See the relevant sections in this debug guide for guidelines about which file(s) to include with the WebCase.

Debug Tools

There are many tools available to address IBERT 7 series GTP transceivers design issues. It is important to know which tools are useful for debugging various situations.

Vivado Lab Tools

Vivado inserts logic analyzer and virtual I/O cores directly into your design. Vivado Lab Tools allows you to set trigger conditions to capture application and integrated block port

signals in hardware. Captured signals can then be analyzed. This feature represents the functionality in the Vivado IDE that is used for logic debugging and validation of a design running in Xilinx FPGA devices in hardware.

The Vivado logic analyzer is used to interact with the logic debug LogiCORE IP cores, including:

- ILA 2.0 (and later versions)
- VIO 2.0 (and later versions)

License Checkers

If the IP requires a license key, the key must be verified. The Vivado design tools have several license check points for gating licensed IP through the flow. If the license check succeeds, the IP can continue generation. Otherwise, generation halts with error. License checkpoints are enforced by the following tools:

- Vivado Synthesis
- Vivado Implementation
- write_bitstream (Tcl command)



IMPORTANT: *IP license level is ignored at checkpoints. The test confirms a valid license exists. It does not check IP license level.*

Additional Resources

Xilinx Resources

For support resources such as Answers, Documentation, Downloads, and Forums, see the Xilinx Support website at:

www.xilinx.com/support.

For a glossary of technical terms used in Xilinx documentation, see:

www.xilinx.com/company/terms.htm.

References

These documents provide supplemental material useful with this product guide:

1. *7 Series FPGAs GTP Transceivers User Guide* ([UG482](#))
 2. *7 Series FPGAs Overview* ([DS180](#))
 3. Vivado™ Design Suite user documentation
 4. *Vivado Design Suite User Guide, Designing with IP* ([UG896](#))
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Revision History

The following table shows the revision history for this document.

Date	Version	Revision
03/20/2013	1.0	Initial Xilinx release of the product guide and replaces DS874.

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